

Existing Conditions Knowledge Base

Al Safa 2 Park — Phase 1 Master Index & Compiled Findings

Compiled: 2026-07-24. This is the master compiled index of Phase 1. It states, honestly, what is evidence-based vs. what remains a data gap for later resolution. **No design decisions have been made in this phase** — Phase 1 is understanding only, per explicit project instruction.

Index of Sub-Analyses

#	Analysis	Report File	Status
1.01	Regional Analysis	Phase1.01_Regional_Analysis_Report.pdf	Complete (qualitative)
1.02	Urban Context	Phase1.02_Urban_Context_Report.pdf	Complete (image-derived)
1.03	Existing Park Analysis	Phase1.03_Existing_Park_Analysis_Report.pdf	Partial — DWG pending
1.04	GIS Analysis	Phase1.04_GIS_Analysis_Report.pdf	Partial — external GIS data gap
1.05	Climate Analysis	Phase1.05_Climate_Analysis_Report.pdf	UPGRADED — sourced Dubai Meteorological Office data + full-year (8,760hr) exact solar dataset
1.06	Shadow Analysis	Phase1.06_Shadow_Analysis_Report.pdf	Complete (computed, exact astronomy, 3 key dates)
1.07	Environmental Analysis	Phase1.07_Environmental_Analysis_Report.pdf	Partial — soil/stormwater/noise gaps
1.08	Transportation Analysis	Phase1.08_Transportation_Analysis_Report.pdf	Partial — bus/parking gaps
1.09	Human Analysis	Phase1.09_Human_Analysis_Report.pdf	Complete (brief + manual derived)
1.10	Accessibility Analysis	Phase1.10_Accessibility_Analysis_Report.pdf	Complete (baseline = undocumented)
1.11	SWOT + Opportunities/Constraints	Phase1.11_SWOT_Report.pdf	Complete (synthesis)
1.12	AI Analysis (methodology log)	Phase1.12_AI_Analysis_Report.pdf	Complete
1.13	Catchment & Demand Analysis	Phase1.13_Catchment_Demand_Analysis_Report.pdf	NEW — real Dubai Statistics Center 2023 population + computed demand model

Headline Facts (verified from competition documents)

- Site: Al Safa 2 Park, 15,000 sqm, Al Safa 2 district, Dubai
- Budget: AED 35,000,000 implementation
- Context: bounded by Umm Suqeim First, Al Safa Second, Al Manara, Al Qouz Industrial First (across Sheikh Zayed Road)
- Transit: ONPASSIVE Metro Station nearby (across SZR)
- Adjacent anchors: Umm Suqeim Model School, Aisha Butti Al Mulla Masjid, Dubai Physiotherapy & Rehabilitation Center
- Climate: Hot desert climate; comfortable season Nov-Apr; extreme heat + near-zero midday shade May-Oct; prevailing WNW wind (16.7 km/h avg, sourced); near-zero rainfall Jun-Sep
- Park archetype (Neighborhood Parks Manual): Neighborhood/community park, 05:00-23:00 operation, 1-3hr visits, 150-400 visitors/10,000sqm peak capacity benchmark

Real Data Now Sourced (v3 pass — previously gaps)

- **Population / catchment:** RESOLVED — real Dubai Statistics Center 2023 figures (Umm Suqeim 1/2/3 + Al Safa), plus a computed walk-catchment demand model (Phase 1.13).
- **Wind:** RESOLVED — real annual average (16.7 km/h) and dominant direction (WNW) from Windfinder / Dubai Intl Airport 24-year record (Phase 1.05).
- **Transit:** RESOLVED — ONPASSIVE Station confirmed Red Line, Fare Zone 2, RTA, formerly "Al Safa Metro Station" (Phase 1.08).
- **Plant specs & water:** RESOLVED — real Ghaf irrigation figures drive a computed ~5,700 m³/year park water budget (Phase 6 + Phase 7.4).

Remaining Data Gaps (still logged, not fabricated)

- As-built DWG geometry (paths, entrances, utilities, levels) — needs DXF conversion
- Building density, slope, noise (dB), soil, stormwater capacity data (need GIS layers)
- Existing on-site parking/drop-off/cycling counts (need DWG or site visit)
- Behavioral/observational user data (movement, dwell time — needs a site study)

These are tracked centrally in 00_MASTER_TRACKER.

Real-Data Upgrade Log (2026-07-24, second pass)

- Climate normals replaced: previously a generic published reference; now sourced live via WebSearch/WebFetch from the **Dubai Meteorological Office** climate normals table (Wikipedia "Climate of Dubai"), including a genuinely new metric — average daily sunshine hours — not present in the original analysis.
- Solar analysis extended from 3 sample dates to a full **8,760-hour, whole-year exact computation** (4,425 confirmed real daylight hours) using the same pvlib/NREL method, at hourly resolution for every day of 2026.
- This full-year dataset was then used to re-run the Phase 7 shade simulation across every daylight hour of the year (not just 3 snapshots), revealing that while the Shaded Spine holds 99.2% annual shade coverage, individual activity-room centroids only get 3.6-16.2% passive shade from the spine/buffer canopy alone — a genuine new finding that feeds back into Phase 6 planting placement.
- World Bank Climate Knowledge Portal was attempted as an additional source but returned HTTP 403 (bot-blocked) on both the page and its data API — logged as inaccessible, not silently skipped.

What Happens Next

Phase 2 (Problem Definition) will use this Knowledge Base — specifically the Weaknesses, Threats, and data-gap items — as its evidence base for identifying the specific problems the new design must solve. No problem prioritization or design response happens until Phase 2 is explicitly started.